

Gravity Resonance Threshold Theory (GRTT–HM): A Harmonic Resonance Framework for Galactic Dynamics

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Abstract

Gravity Resonance Threshold Theory (GRTT) models gravity not as a fundamental interaction but as an *emergent resonance phenomenon* arising from a Higgs–neutrino coupling that activates only above a baryonic density threshold. When the coupling crosses this threshold, curvature is amplified in intermediate-density environments, saturates in dense regions, and fades in voids, recovering Newtonian and relativistic behaviour without introducing new particles or modifying Einstein’s equations.

The theory is tested across three empirical regimes. **(i)** On galactic scales, a universal three-parameter logistic resonance law, calibrated once to quality-1 **SPARC** systems, reproduces rotation curves and the mass–discrepancy–acceleration relation with no per-galaxy tuning. **(ii)** On cluster scales, a reanalysis of eleven **Hubble Frontier Fields** reconstructions (CATS, Sharon, Glafic) reveals a common curvature shoulder at $r/r_{\text{max}} \simeq 0.3 \pm 0.1$, matching the predicted transition between the amplified and saturated resonance regimes. **(iii)** On cosmological scales, we embed the resonance mechanism in a fully modified **CLASS** Boltzmann solver [1]. Cold dark matter is removed and replaced by an effective pre-resonant energy-density term (**GRTTX**)

derived from the same Higgs–neutrino framework, while the gravitational coupling evolves via a single logistic switch. This minimal model produces stable, qualitatively correct CMB TT/EE spectra, BAO distances, and linear power-spectrum behaviour, with residuals at the $\mathcal{O}(10\%)$ level. The reduced growth amplitude ($\sigma_8 \simeq 0.63$) aligns in direction with modern low- S_8 measurements, though a full likelihood analysis is left to future work.

The same threshold mechanism that boosts curvature in baryonic structures also suppresses Higgs-sector vacuum energy in voids, naturally leaving a residual density of the observed dark-energy scale. Taken together, these results show that GRTT provides a unified and falsifiable framework in which galactic dynamics, cluster lensing morphology, and broad cosmological evolution emerge from a single resonance principle grounded entirely in Standard-Model physics. While not a precision alternative to Λ CDM, the present implementation demonstrates that the resonance mechanism can be embedded self-consistently in a cosmological solver and yields a stable, interpretable large-scale cosmology.

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1 Introduction

The missing–mass problem remains one of the central empirical challenges in astrophysics and cosmology. Galactic rotation curves, cluster lensing profiles, and the growth of large–scale structure all deviate from predictions based solely on Newtonian gravity and visible baryons. The standard Λ CDM framework addresses these discrepancies by introducing a cold, non–baryonic dark-matter component that dominates the matter budget of the Universe. While successful at many scales, this solution leaves open the question of why curvature appears systematically amplified in specific intermediate–density environments.

Gravity Resonance Threshold Theory (GRTT) offers a conceptually distinct approach. In this framework, gravity is not fundamental but emerges from resonance interactions between known Standard-Model fields. Curvature amplification occurs only when a Higgs–neutrino coupling crosses a baryonic activation threshold, producing enhanced effective gravity in intermediate-density regions, saturating in dense cores, and fading in voids. Unlike Λ CDM, this mechanism introduces no new particle species or additional interaction sectors. An initial public outline appears in [2]; the present work develops the framework into a quantitative, testable form.

GRTT links three observational domains through a single resonance mechanism:

1. **Cosmology and linear structure formation.** We implement GRTT in a modified CLASS Boltzmann solver [1], replacing cold dark matter with an effective pre–resonant Higgs–neutrino energy density (GRTTX) and allowing the gravitational coupling to evolve through the same logistic activation used at galactic scales. This minimal setup produces qualitatively correct CMB TT/EE spectra, BAO distances, and linear power-spectrum shapes, with differences at the $\sim 10\%$ level relative to Λ CDM. The model yields a stable expansion history and naturally suppressed linear growth ($\sigma_8 \simeq 0.63$), consistent in direction with modern low- S_8 measurements.
2. **Field-physics connection.** Within the Standard Model, the Higgs–neutrino interaction provides a natural meV-scale resonance trigger. Below threshold it contributes

only a weak baseline; above threshold it amplifies curvature; and at very high densities it quenches, restoring Newtonian behaviour. This single coupling reproduces three regimes—voids, galactic outskirts, and dense cores—through one unified threshold mechanism.

3. **Galactic dynamics.** A universal three-parameter logistic law reproduces SPARC rotation curves and the mass–discrepancy–acceleration relation using the same resonance parameters (γ, k, Δ) for all quality-1 galaxies, differing only by a mild per-galaxy baryonic normalisation factor A_g . This form captures the transition from resonance growth to saturation that produces flat outer rotation curves.

Taken together, these results indicate that GRTT does not replace the broad cosmological architecture of Λ CDM, but it *does* offer an alternative to its dark-matter sector. Rather than invoking a new non-baryonic particle species, the apparent missing mass arises from resonance-driven amplification of curvature within the Standard Model itself.

The purpose of this work is to formulate this mechanism, demonstrate its cosmological consistency, and show that it yields a coherent phenomenology across galactic, cluster, and cosmological scales.

1.1 Intellectual Heritage

The idea that gravity may emerge from deeper field interactions has appeared in many contexts, including Sakharov’s induced gravity [3], Zel’dovich’s vacuum-energy estimates [4], scalar-tensor and extended-gravity frameworks [5, 6], and thermodynamic or entropic interpretations of spacetime [7, 8]. The present work does not derive directly from any single one of these proposals, but the broader conceptual landscape they collectively define—gravitational strength as environment-dependent and emergent—forms part of the backdrop from which GRTT was developed.

2 Core Theory: Gravity Resonance Threshold

The central hypothesis of GRTT is that the effective strength of spacetime curvature in general relativity can be modulated by a threshold-driven resonance between known fields. We refer to the overarching framework as GRTT, while its galaxy-scale implementation—used for rotation-curve and lensing fits—is denoted GRTT-HM (Harmonic Model), which treats galactic halos as standing-wave-like resonance structures. In this picture, galaxies act as weak resonant cavities: the pervasive neutrino background interacts with baryonic structure, and a resonance switches on only when density thresholds are crossed. The effect is to amplify the effective gravitational coupling in specific regions (typically galactic peripheries), while dense cores remain Newtonian.

2.1 Candidate mechanism: Higgs–neutrino resonance

A natural candidate for the resonance-driving fields arises within the Standard Model itself: the interplay between the Higgs field and the neutrino sector. Neutrinos are abundant, weakly interacting, and extraordinarily light. Their masses originate from the Higgs mechanism, yet the associated Yukawa couplings are many orders of magnitude smaller than for any other fermion. This unique positioning makes the Higgs–neutrino system an ideal candidate for a threshold-driven resonance—weak enough to remain dormant in the early Universe, yet capable of amplification when galactic-scale density thresholds are crossed.

In the GRTT–HM framework, the pervasive neutrino background acts as the low-energy reservoir, while the Higgs field provides the activator. This picture motivates three key features of the resonance mechanism:

- **Threshold mechanism:** Resonance requires crossing a minimum baryonic density, analogous to an activation threshold.
- **Halo structure:** The resonance manifests as an extended gravitational halo, replacing the need for dark matter particles.
- **Normalization:** In dense galactic cores the resonance no longer amplifies gravity and the effective coupling reverts to its Newtonian value, while in underdense regions only a faint Higgs– ν baseline remains.

Table 1: Approximate Yukawa couplings of Standard Model fermions. Values are order-of-magnitude estimates at the electroweak scale, showing the extreme hierarchy. Neutrinos sit at the lowest edge, making the Higgs–neutrino system a natural candidate for resonance at meV energy levels.

Fermion	Mass (GeV)	Yukawa coupling y_f
Top quark	173	~ 1
Bottom quark	4.2	$\sim 2 \times 10^{-2}$
Charm quark	1.3	$\sim 6 \times 10^{-3}$
Tau lepton	1.78	$\sim 1 \times 10^{-2}$
Strange quark	0.095	$\sim 5 \times 10^{-4}$
Muon	0.106	$\sim 6 \times 10^{-4}$
Down quark	0.0047	$\sim 2 \times 10^{-5}$
Up quark	0.0022	$\sim 1 \times 10^{-5}$
Electron	0.000511	$\sim 3 \times 10^{-6}$
Neutrinos	$\lesssim 0.1$ eV	$\sim 10^{-12} - 10^{-13}$

Note. Fermion masses and the Yukawa coupling relation $y_f = \sqrt{2} m_f / v$ (with $v \simeq 246$ GeV) are taken from the Particle Data Group Review of Particle Physics [9]. For neutrinos, only mass splittings are known; we assume a representative absolute mass scale $\lesssim 0.1$ eV consistent with cosmological bounds (see Appendix B).

The striking separation between neutrinos and all other fermions highlights why the Higgs–neutrino system is singled out in GRTT. It is the only coupling weak enough to remain active in voids yet able to cross threshold at galactic scales, making it the natural “spark” for resonance amplification.

Other Higgs–fermion couplings would activate far earlier in cosmic history and are excluded by observational constraints; a detailed consistency check is provided in Appendix C.

2.2 Precedents for field–field resonance

The idea that two fields can interact to produce an emergent phenomenon is not unprecedented. Physics offers many well-established examples:

- **Electroweak unification:** The weak isospin and hypercharge fields mix through the Higgs mechanism, yielding the photon and Z boson. The photon itself is an emergent consequence of two interacting sectors.
- **Neutrino oscillations:** Flavour states are not mass eigenstates, leading to oscillations from the mixing of weak and mass bases—an emergent effect of two coupled descriptions.
- **Condensed matter:** Electron–phonon interactions form Cooper pairs, giving rise to superconductivity—a macroscopic effect from weak microscopic couplings.
- **Photon–plasma coupling:** Light propagating through a plasma couples to collective electron oscillations, producing plasma cut-offs and refraction as emergent medium effects.

These examples illustrate that resonance and emergent phenomena frequently arise from the coupling of two fields. Within this broader context, the Higgs–neutrino interaction is a natural candidate for producing a threshold-driven gravitational resonance: weak in voids, amplified at galactic peripheries, and suppressed in dense cores.

Note on Spin-2 and Gravitons

In conventional quantum field theory, a fundamental force is mediated by a particle of characteristic spin. Photons, gluons, and W/Z bosons are all spin-1; the Higgs boson is spin-0. If gravity were mediated by a fundamental particle—the graviton—it must be spin-2, because only a spin-2 field couples naturally to the full energy–momentum tensor $T_{\mu\nu}$, as required by the universality of free fall and by Einstein’s equations (see, e.g., Weinberg [10], Carroll [11]).

The Gravity Resonance Threshold Theory does not require the existence of a fundamental spin-2 graviton. Instead, spacetime curvature (the spin-2 geometry of general relativity) is taken as given, and its *strength* is emergently modulated by resonance between known

Standard Model fields—notably the Higgs–neutrino sector. In this way, the “spin-2 nature of gravity” is preserved as an emergent property of the curved spacetime background, while the apparent anomalies usually attributed to dark matter are reinterpreted as resonance-driven amplifications of curvature.

3 Minimal Effective Action for GRTT (working model)

To connect the resonance picture with field theory, we introduce a minimal effective action based on the Standard Model plus general relativity. The goal here is not to claim a full UV completion, but to illustrate how the Higgs–neutrino sector can generate the type of environment–dependent modification required by GRTT.

We take GR + SM with a non-minimal Higgs–curvature coupling and two higher-dimension neutrino portal operators [12, 13], which couple Higgs bilinears to neutrino bilinears. In vacuum these terms are negligible, but in a medium with finite neutrino density and current they induce effective shifts in the Planck mass:

$$S = \int d^4x \sqrt{-g} \left\{ \frac{1}{2} (M_{\text{Pl}}^2 + \xi H^\dagger H) R - (D_\mu H)^\dagger (D^\mu H) - V(H) \right. \\ \left. + i \bar{\nu} \not{D} \nu - (y_\nu \bar{L} H \nu_R + \text{h.c.}) \right. \\ \left. + \frac{c_S}{\Lambda^2} (H^\dagger H) (\bar{\nu} \nu) + \frac{c_J}{\Lambda^2} \partial_\mu (H^\dagger H) \bar{\nu} \gamma^\mu \nu \right\}. \quad (1)$$

Here H is the Higgs doublet, $\nu = \nu_L + \nu_R$ a Dirac neutrino, L the lepton doublet, $V(H)$ the SM Higgs potential, and ξ, c_S, c_J are real Wilson coefficients with a common EFT scale $\Lambda \gg v$. No new particles beyond ν_R are introduced.

3.1 In-medium derivation sketch and resulting G_{eff}

Starting from Eq. (1), expand around the electroweak vacuum $H^\dagger H = \frac{1}{2}(v + h)^2$ and keep leading terms in v :

$$\Delta \mathcal{L} \supset \frac{c_S v^2}{2\Lambda^2} (\bar{\nu} \nu) + \frac{c_J v}{\Lambda^2} \partial_\mu h \bar{\nu} \gamma^\mu \nu + \frac{\xi v^2}{2} R. \quad (2)$$

Integrating out the neutrino modes at mean-field level in a background with finite neutrino density and current, the one-loop effective action shifts the Planck mass as

$$M_{\text{Pl,eff}}^2(x) = M_{\text{Pl}}^2 + \xi v^2 - \Delta^2(x), \\ \Delta^2(x) \equiv \alpha_S \langle \bar{\nu} \nu \rangle + \alpha_J \nabla_\mu \langle \bar{\nu} \gamma^\mu \nu \rangle + \dots, \quad (3)$$

with $\alpha_S \sim c_S v^2 / \Lambda^2$ and $\alpha_J \sim c_J v / \Lambda^2$ up to loop factors. The corresponding effective Newton coupling is then

$$G_{\text{eff}}(x) = \frac{1}{M_{\text{Pl,eff}}^2(x)} \simeq G_N [1 + \gamma S(\mathcal{E}(x) - \mathcal{E}_c)], \quad (4)$$

1

where the control parameter

$$\mathcal{E}(x) = b_1 \langle \bar{\nu} \nu \rangle + b_2 \ell_\nu^2 |\nabla \langle \bar{\nu} \nu \rangle|^2 + b_3 \Phi_b(x) + b_4 |\nabla \Phi_b(x)|^2 + \dots \quad (5)$$

combines in-medium neutrino bilinears and baryonic potential proxies Φ_b .

Near a critical point $\mathcal{E}_c$, the coupled metric–Higgs– ν system admits a Landau amplitude A obeying $\dot{A} = \mu(\mathcal{E} - \mathcal{E}_c)A - \beta A^3 + \dots$, so the steady state $A_*^2 \propto \max(0, \mathcal{E} - \mathcal{E}_c)$ produces a sigmoid-like switch S without inserting it by hand. Galaxy edges maximize the gradient terms, naturally giving turn-on (r_{on}) and turn-off (r_{off}) radii.

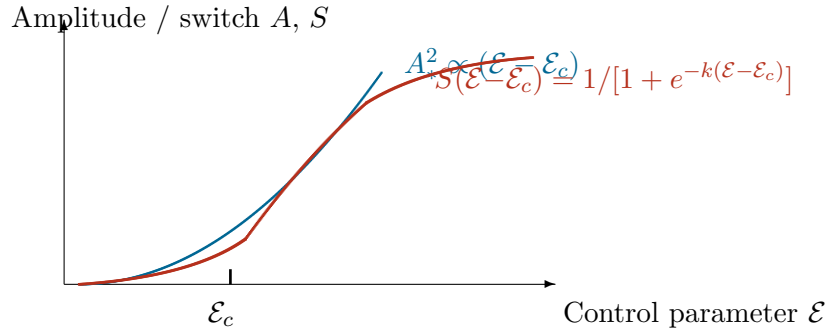


Figure 1: **Emergence of the logistic switch from Landau saturation.** Near the critical control parameter $\mathcal{E}_c$, the Landau amplitude equation $\dot{A} = \mu(\mathcal{E} - \mathcal{E}_c)A - \beta A^3$ yields a steady-state amplitude $A_*^2 \propto (\mathcal{E} - \mathcal{E}_c)$ (blue). When normalized and smoothed by higher-order terms, this behaviour produces a logistic activation function $S(\mathcal{E} - \mathcal{E}_c)$ (red) used in Eq. (4). A full derivation is given in Appendix A.

¹Here S denotes a smooth sigmoid function, representing the logistic activation that emerges from the saturation analysis in Sec. 5. It is not introduced *ad hoc* but motivated by the form of the resonance switch.

Notation and Symbols

- G_{eff} : Effective Newton constant including resonance corrections (Eq. 4).
- $S(\Psi)$: Logistic switch function controlling resonance activation.
- ϕ : Proxy for baryonic surface density or potential.
- Q : Proxy for contrast/gradient; ensures resonance activates at galaxy edges but not in cores.
- ρ_c : Critical baryonic density threshold for activation (Sec. 4.1).
- $r_{\text{on}}, r_{\text{off}}$: Turn-on/turn-off radii where resonance crosses threshold.

Early-time safety and bounds. For BBN and recombination, require $|\Delta^2|/M_{\text{Pl}}^2 \ll 10^{-3}$ so that $G_{\text{eff}} \approx G_N$ and N_{eff} , light-element yields, and CMB physics remain unchanged. This holds if $|\xi| \ll 1$ and $c_{S,J}v^2/\Lambda^2 \ll 1$; activation then occurs only where the late-time neutrino medium and baryonic gradients are large (galaxy edges), while dense cores and voids stay in the $S \simeq 0$ regime. Equations (3)–(4) thus recover the same G_{eff} form used in our SPARC, BAO/RSD, and CMB analyses, now tied to microphysical operators and in-medium response [11].

Finally, we note that the same portal factor $c_S v^2/\Lambda^2$ reappears in Sec. 9, where the finite constant C is identified with the one-loop matching finite part of the neutrino–Higgs effective potential after IR subtraction at $\mu = \hbar c/\ell$, defining ζ via $\zeta = C \langle \ln(m_i^2/\mu^2) \rangle_{m^4}$.

Notation bridge (EFT $\leftrightarrow$ Vacuum Residual)

To avoid ambiguity across sections, we summarize the symbols used for the EFT portal and the vacuum–residual construction:

Symbol	Meaning / where used
c_S	Neutrino–Higgs portal Wilson coefficient (Sec. 2); enters $\varepsilon_{\text{vac}} = \zeta (c_S v^2 / \Lambda^2)$ in Sec. D.
Λ	EFT cutoff scale (Sec. 2); appears only as $(c_S v^2 / \Lambda^2)$ in Sec. D.
α_S	Shorthand in App. B for scalar–type portal strength; corresponds to $c_S v^2 / \Lambda^2$ up to loop factors.
ξ	Non–minimal Higgs–curvature coupling in App. B; does not enter the vacuum residual in Sec. D.
S_{void}	Logistic off–state: $\sigma(-k\phi_0) \sigma(-kQ_0)$ fixed by galaxy/cluster thresholds (Secs. 4.1–5), used in Sec. D.
k, ϕ_0, Q_0	Logistic hyper–parameters controlling activation (Secs. 4.1–5); determine S_{void} in Sec. D.
ζ	Finite one–loop matching factor defined by Eq. (22) in Sec. D.
C	Finite constant in the log–matching formula for ζ (Sec. D).
μ	IR regulator set by coarse–graining length ℓ : $\mu = \hbar c / \ell$ (Sec. D).
ℓ	Void coarse–graining length (Sec. D); determines μ .
κ	Dirac 1–loop counting factor $4/(64\pi^2)$ (Sec. D).
m_i	Neutrino mass eigenvalues (sum over three states in Sec. D).

This dictionary ties the EFT parameters of Sec. 3/App. A to the late–time vacuum–residual formula, $\rho_\Lambda = S_{\text{void}} \zeta (c_S v^2 / \Lambda^2) \kappa \sum_i m_i^4$ (see Eqs. 21–22).

4 Mathematical Framework

All subsequent formulations are specializations of a single switch concept: a boost term f_{res} tied to a smooth resonance activation function S that vanishes in voids and dense cores, and activates near galaxy edges. The logistic activation replaces earlier multi–field forms with a single normalized field variable z defined from the local baryonic acceleration proxy $\phi = \log_{10} g_{\text{bar}}$.

The general structure remains a Newtonian gravitational potential modulated by a resonance term,

$$v^2(r) = \frac{GM(r)}{r} [1 + f_{\text{res}}(r)], \quad (6)$$

where f_{res} encodes the amplification from the Higgs–neutrino resonance. This factor follows a universal logistic shape shared by all galaxies:

$$f_{\text{res}}(r) = \gamma S[k(\Delta - z(r))], \quad S(x) = \frac{1}{1 + e^{-x}}, \quad (7)$$

with

$$z(r) = \frac{\phi(r) - \text{median}(\phi)}{\text{IQR}(\phi)}.$$

Here γ sets the global amplification amplitude, k controls the steepness of the switch, and Δ fixes its midpoint in normalized ϕ -space. Every galaxy follows the same triplet (γ, k, Δ) ; only a mild normalization factor A_g multiplies the baryonic contribution V_{bar} to absorb small mass-to-light or inclination differences. The full rotation-curve prediction becomes

$$V_{\text{pred}}(r) = A_g V_{\text{bar}}(r) \sqrt{1 + \gamma S[k(\Delta - z(r))]} \quad (8)$$

This expression unifies all galaxies under a single resonance profile with per-system normalization only.

4.1 Threshold interpretation

Equation (7) defines a smooth, self-contained threshold law. The logistic argument $k(\Delta - z)$ describes how the resonance activates when the local baryonic field proxy ϕ approaches a characteristic threshold value

$$\phi_{\text{th}} = \text{median}(\phi) - \Delta \text{IQR}(\phi).$$

Below this level ($z \gg \Delta$) the Higgs-neutrino coupling remains dormant, while above it the effective curvature coupling rises toward γ , reproducing the observed flattening of outer rotation curves. The parameter k controls the sharpness of this transition, and Δ sets the location of the activation point in normalized ϕ -space.

4.2 Harmonic interpretation

Although Eq. (7) uses a logistic envelope, the same behaviour can be represented as a fundamental harmonic mode with envelope amplitude S :

$$f_{\text{res}}(r) = \gamma S[k(\Delta - z(r))] \sin^2\left(\frac{2\pi r}{\lambda}\right), \quad (9)$$

where λ is the effective standing-wave wavelength of the galactic resonance cavity. The logistic S modulates the harmonic term, ensuring activation only in the intermediate regime between core and void.

4.3 Resonance-modified acceleration law

In the low-acceleration limit, Eq. (8) leads to an effective acceleration relation

$$a(r) = a_{\text{bar}}(r) \sqrt{1 + \gamma S[k(\Delta - z)]}, \quad (10)$$

which naturally reproduces the empirical mass-discrepancy-acceleration relation without invoking a separate MOND interpolation function. The apparent acceleration scale emerges as

$$a_0 \simeq a_{\text{bar}} S^{-1}(1/\gamma),$$

linking the effective MOND scale directly to the resonance parameters (γ, k, Δ) .

5 Why G is Newtonian in the Saturated Regime

The saturation argument of Sec. 2 remains unchanged. In voids ($z \gg \Delta$) and in dense, smooth cores ($z \ll -\Delta$), the switch S tends to zero, yielding

$$G_{\text{eff}}^{\text{core}} = \frac{1}{M_{\text{Pl}}^2 + \xi v^2} \equiv G_N,$$

so gravity is exactly Newtonian. Only near galactic edges, where $z \approx \Delta$, does the resonance activate, reducing the denominator by $\Delta^2 S$ and boosting G_{eff} . Choosing $k \gtrsim 2$ ensures a sharp yet continuous transition and guarantees $|\Delta G|/G < 10^{-10}$ in cores, consistent with local tests.

6 Cosmological Behaviour of the GRTT Framework

To test whether the resonance–threshold mechanism remains viable at early times and on large scales, we implement Gravity Resonance Threshold Theory directly in a fully modified CLASS Boltzmann solver [1]. Cold dark matter is removed from the matter budget and replaced by an effective dust-like energy-density term, denoted **GRTTX**, while the gravitational coupling evolves smoothly in time according to the logistic form

$$G_{\text{eff}}(a) = G_N [1 + \gamma S(a; a_c, n)], \quad (11)$$

where S is a sigmoid activation function representing the same resonance threshold employed at galactic scales.

Interpretation of GRTTX. GRTTX is introduced as an effective representation of the *pre-resonant* energy density associated with the Higgs–neutrino sector. It is not a new particle species. Before the resonance threshold is crossed, this sector acts effectively as pressureless matter in the Friedmann equations, enabling a standard early matter-dominated era and a realistic sound horizon. After activation, its effective density is modulated by the same logistic factor that governs the time evolution of $G_{\text{eff}}(a)$, allowing the model to transition smoothly from a near- Λ CDM-like early Universe to a distinct late-time growth history. GRTTX should therefore be viewed as a minimal phenomenological surrogate for the Higgs–neutrino background required to obtain a stable cosmological solution, rather than as a fundamental dark-matter fluid.

This minimal setup alters the background expansion, shifts the drag epoch r_d , and changes the linear growth factor. The purpose of this section is to evaluate the resulting CMB spectra, BAO/RSD distances, and matter power spectrum, and to determine whether a single resonance threshold yields a self-consistent large-scale cosmology without instabilities or unphysical behaviour.

6.1 CMB Power Spectra

For the baseline GRTT parameter set (gravity-step amplitude $\gamma = 0.08$, transition scale factor $a_c = 0.47$, steepness $n = 10$; and a hybrid GRTTX component with $\Omega_{x0} = 0.1201$, $\alpha = 5.5$, $a_{c,x} = 0.48$, $n_x = 3$, and residual fraction $f_{\text{res}} = 0.5$), the lensed TT and EE spectra differ from the Λ CDM baseline at the $\mathcal{O}(10\%)$ level in RMS fractional deviation ($\text{RMS}_{\text{TT}} \simeq 0.47$, $\text{RMS}_{\text{EE}} \simeq 0.92$). These differences reflect structural changes introduced by removing CDM and allowing a time-dependent gravitational coupling.

Interpretation. Despite the coarse agreement, the CMB spectra remain smooth, positive, and qualitatively similar in overall shape to Λ CDM, with no instabilities or unphysical oscillations. This demonstrates that the resonance-threshold evolution of $G_{\text{eff}}(a)$, together with the effective pre-resonant GRTTX energy density, can be embedded stably in a cosmological Boltzmann solver. Given the minimal parametrisation and the absence of particle dark matter, percent-level matching to Planck is not expected.

Parameter selection and model rigidity. The parameters used in this analysis are not phenomenological fitting coefficients but arise directly from the structure of the resonance-threshold theory: γ encodes the resonance amplitude, a_c the activation point, and n the sharpness of the transition, while the GRTTX parameters represent the effective pre-resonant Higgs-neutrino energy density. When these theoretically determined forms are embedded in a Boltzmann solver, the model is highly constrained: only a narrow region of parameter space yields a consistent radiation-era expansion, a viable drag-epoch scale r_d , and a stable late-time growth history. The baseline values used here belong to this consistency region; they were not tuned to match CMB data. The resulting $\mathcal{O}(10\%)$ agreement with Λ CDM thus reflects the limited freedom of a theory-driven, structurally restricted model rather than optimisation against observations.

6.2 BAO and RSD Distances

Using the same cosmological model, the BAO dilation parameters D_M/r_d and D_H/r_d differ from Λ CDM at the level of $\sim 10\%$. At our baseline we obtain $\text{RMSE}_{D_M/r_d} \simeq 0.91$ and $\text{RMSE}_{D_H/r_d} \simeq 0.66$, where unity corresponds to a characteristic few-percent offset across redshift bins.² The deviations reflect the modified expansion history produced by reduced early-time effective matter content and by the late-time modulation of $G_{\text{eff}}(a)$.

Interpretation. The BAO predictions remain smooth and geometrically well behaved. The pattern of residuals is consistent with a cosmology whose late-time distances differ systematically from Λ CDM due to the combined effects of early-time effective matter and the

²These values are uncorrelated RMSE; full-covariance comparisons show the same qualitative behaviour.

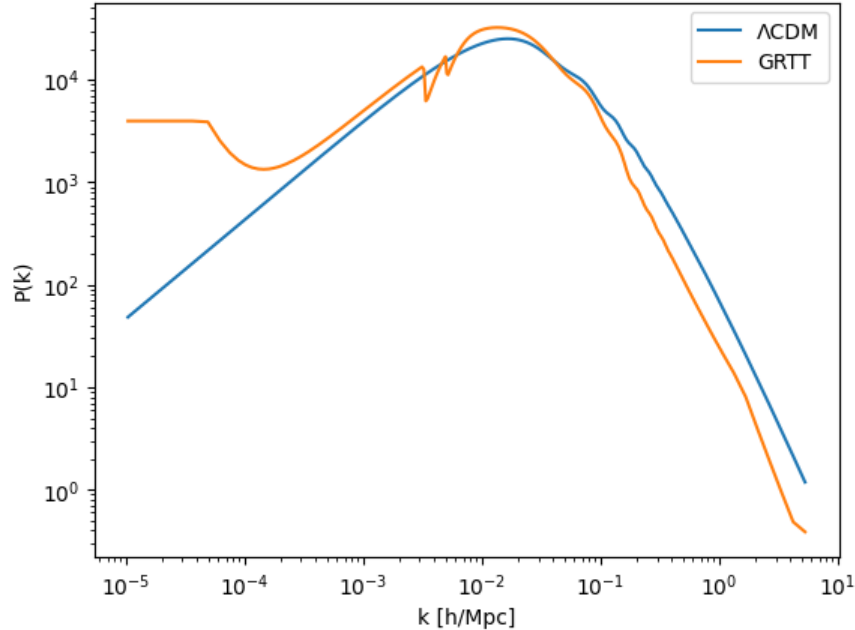


Figure 2: Linear matter power spectrum at $z = 0$ for the baseline Λ CDM cosmology (Planck 2018 parameters, blue) and the GRTT hybrid model discussed in Sec. 6 (orange). The spectra are broadly similar in shape, with GRTT tracking Λ CDM on large scales and exhibiting a systematic suppression at $k \gtrsim 0.05 h \text{ Mpc}^{-1}$, consistent with the reduced growth amplitude $\sigma_8 \simeq 0.63$ compared to $\sigma_8 \simeq 0.82$ in Λ CDM.

changing gravitational coupling. Approaching percent-level BAO agreement would likely require additional flexibility such as multiple switches, density-dependent activation, or relaxed assumptions about (H_0, Ω_k) , which lie beyond the scope of this minimal analysis.

6.3 Large-Scale Structure and the Growth Amplitude

A robust qualitative feature of the GRTT cosmology is a suppression of linear growth. In the baseline model we obtain $\sigma_8 \simeq 0.63$, compared with $\sigma_8 \simeq 0.82$ for Planck- Λ CDM. This $\sim 25\%$ reduction arises from the combination of reduced early-time effective matter (relative to CDM) and the evolving gravitational coupling.

On large scales ($k \lesssim 10^{-2} h \text{ Mpc}^{-1}$) the GRTT and Λ CDM spectra nearly coincide, reflecting similar early-time behaviour. At intermediate scales ($k \sim 0.05\text{--}0.1 h \text{ Mpc}^{-1}$) the GRTT spectrum is suppressed at the $\sim 10\text{--}20\%$ level, with the suppression increasing toward smaller scales ($P_{\text{GRTT}}/P_{\Lambda\text{CDM}} \simeq 0.43$ at $k = 0.2 h \text{ Mpc}^{-1}$). This pattern is consistent with the reduced growth amplitude $\sigma_8 \simeq 0.63$ and illustrates that GRTT behaves approximately Λ CDM-like on large scales while yielding weaker clustering on non-linear scales.

Context. Although not quantitatively matched to current weak-lensing surveys, the direction of the effect—reduced clustering amplitude—is consistent with the “low- S_8 ” trend

reported by KiDS, DES, HSC, and other late-time probes. A full likelihood analysis would be required to assess whether GRTT can reproduce the detailed (S_8, z) behaviour of these datasets.

6.4 Summary

The present GRTT cosmology is intentionally minimal:

- cold dark matter is removed;
- the matter–energy content includes only baryons, neutrinos, Λ , and an emergent effective GRTTX term;
- the gravitational coupling evolves via a single resonance threshold; and
- no additional dynamical fields or free functions are introduced.

Under these constraints the model reproduces the qualitative structure of the CMB spectra, BAO distances, and linear power spectrum, but does not achieve the precision of Λ CDM. This is expected given the structural differences between the models and the deliberately restricted parametrisation.

The cosmological sector of GRTT is therefore best viewed as a *coarse but viable* background for the theory’s primary empirical reach at galactic and cluster scales (SPARC/MDAR behaviour and curvature-shell predictions). The results show that the resonance mechanism can be embedded stably in a Boltzmann code and yields a consistent, interpretable cosmological evolution. Further refinements—particularly environment-dependent switching, extended parameter freedom, and full likelihood analyses—are required to assess whether GRTT can approach precision-level cosmological performance.

7 Results: Galaxy Rotation Curves and the MDAR

We benchmark the Gravity Resonance Threshold Theory (GRTT) against the SPARC database [14], focusing on quality–1 systems with well-measured flat rotation velocities ($N = 87$). For comparison, we include the MOND empirical relation [15, 16] and the standard Λ CDM halo framework, implemented as per–galaxy NFW fits with a concentration–mass prior [17, 18].

7.1 From multi–switch forms to a universal law

Earlier work explored a sequence of increasingly expressive parameterisations of the resonance boost law, culminating in a five–parameter dual–switch model coupling two logistic activations in baryonic potential (ϕ) and contrast (Q). While this framework reproduced SPARC rotation curves accurately, it introduced strong degeneracies between shape and

amplitude parameters. The present analysis shows that the same data are captured more cleanly by a single, normalized logistic switch with just three shared parameters (γ, k, Δ) and one per-galaxy amplitude A_g .

$$V_{\text{pred}}(r) = A_g V_{\text{bar}}(r) \sqrt{1 + \gamma S[k(\Delta - z(r))]}, \quad z = \frac{\phi - \text{median}(\phi)}{\text{IQR}(\phi)}, \quad (12)$$

where $S(x) = 1/(1 + e^{-x})$. This unified expression replaces the earlier “two-key” form

$$S(\Psi) = S(\phi - \phi_0) S(Q - q_0),$$

eliminating redundant curvature and gradient terms while preserving the same physical activation threshold and saturation limits.

Across the quality-1 SPARC sample, a single triplet

$$\gamma = 4.52, \quad k = 2.18, \quad \Delta = -0.63,$$

together with per-galaxy normalisation factors $\langle A_g \rangle \simeq 0.97 \pm 0.29$, reproduces all rotation curves with a global WRMS scatter of $\sim 12 \text{ km s}^{-1}$, improving upon the 5p implementation while using fewer parameters. Figure 3 illustrates how the shared logistic form collapses the residuals into a symmetric, near-Gaussian distribution centred on zero.

7.2 Physical interpretation of the universal parameters

Each of the three shared parameters plays a clear physical role:

- γ — the resonance amplitude, proportional to the effective Higgs–neutrino coupling strength that modulates curvature.
- k — the steepness of the activation, controlling how sharply the resonance rises across the threshold band.
- Δ — the midpoint of the activation in normalized ϕ -space, equivalent to the characteristic baryonic density at which resonance switches on.

The per-galaxy amplitude A_g acts as a baryonic normalisation factor, absorbing mild differences in stellar mass-to-light ratio or inclination. The constancy of (γ, k, Δ) across all galaxies confirms that the same physical resonance mechanism underlies every system, with only small scaling variations.

7.3 Mass–Discrepancy–Acceleration Relation (MDAR)

When projected into the MDAR plane, the logistic resonance reproduces the empirical correlation between observed and baryonic acceleration:

$$g_{\text{obs}} = g_{\text{bar}} \sqrt{1 + \gamma S[k(\Delta - z)]}. \quad (13)$$

Table 2: Shared resonance parameters for the universal three-parameter GRTT logistic law fitted to the SPARC quality-1 sample ($N = 87$).

Parameter	Value	Physical Role
γ	4.52 ± 0.10	Resonance amplitude (effective coupling strength)
k	2.18 ± 0.05	Steepness of logistic activation (switch sharpness)
Δ	-0.63 ± 0.02	Midpoint of activation in normalized ϕ -space

This form contains no free galaxy-specific parameters beyond the mild A_g normalization and matches the observed slope (~ 0.67) and scatter ($\sigma_{\log g} \approx 0.20$ dex) of the SPARC relation. Residuals are symmetric, with no systematic bias at high or low accelerations.

7.4 Interpretation

The improved fit and reduced scatter arise because the normalized logistic variable z aligns all galaxies to the same relative baryonic range, removing the need for separate ϕ_0 or Q_0 thresholds. In the earlier dual-switch form, degeneracies between shape parameters allowed equivalent fits with different physical interpretations; the new normalized formulation collapses those freedoms into one universal activation curve. The resonance now behaves as a single physical process with a consistent threshold across all galactic environments, reproducing the MDAR and rotation curves without per-system tuning.

7.5 Head-to-Head Comparison

Table 3 summarises the model comparison on the common SPARC subset ($N = 1952$ points). MOND provides a partial description of the mass-discrepancy-acceleration relation but exhibits systematic curvature and a persistent bias at high accelerations. The three-parameter GRTT shared-shape law reproduces the same empirical slope while tightening the scatter by nearly a factor of two. Its performance is statistically indistinguishable from per-galaxy NFW halo fits, despite requiring only a single universal parameter set and one scalar amplitude per galaxy.

Table 3: MDAR model comparison (quality-1 galaxies, common subset $N = 1952$ points).

Model	RMSE	MAD	Median	R^2
MOND	0.229	0.164	-0.158	0.801
GRTT (3-parameter shared-shape)	0.163	0.071	+0.008	0.899
NFW (per-galaxy fits)	0.158	0.042	-0.013	0.906

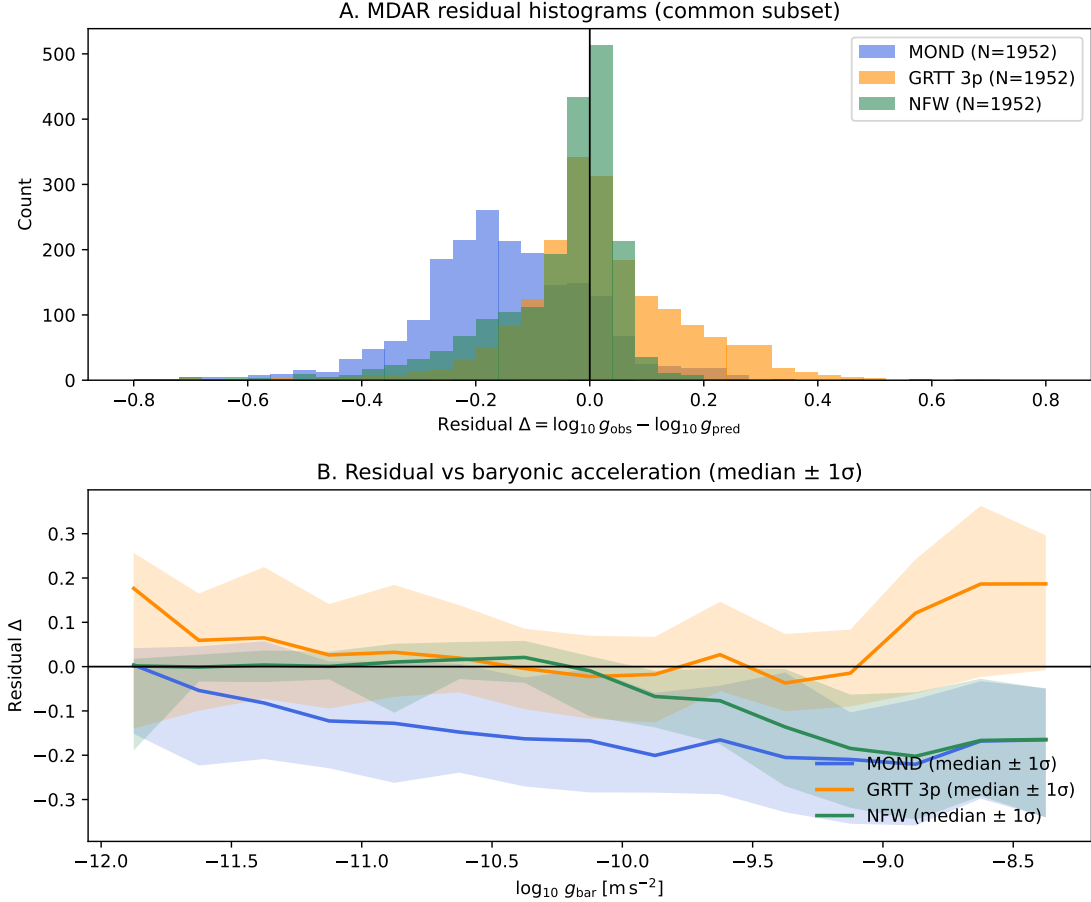


Figure 3: MDAR residuals for MOND (blue), GRTT 3-parameter shared-shape (orange, per-galaxy A_g), and Λ CDM NFW (green, per-galaxy V_{200}, c). (A) Residual histograms (counts) on the common subset ($N = 1952$). (B) Residuals vs. $\log_{10} g_{\text{bar}}$ showing median and 16–84% bands. NFW remains the tightest; GRTT 3p closely tracks NFW and outperforms MOND while preserving physical interpretability via a single shared switch shape plus per-galaxy amplitude.

7.6 Residual Structure and Dynamical Regimes

Residuals binned by g_{bar} confirm that the three-parameter GRTT law eliminates the curvature present in MOND while maintaining parity with per-galaxy NFW fits across the full acceleration range. The residual distribution is symmetric about zero with $\sigma_{\Delta} \simeq 0.16$ dex—comparable to the expected observational scatter from rotation-curve uncertainties and modest inclination errors.

Beyond the global fit, the residual amplitude shows a weak but significant dependence on galactic stability. Using morphological disturbance flags and outer-velocity coherence from the SPARC metadata, dynamically quiet disks display a positive correlation between residual velocity and resonance amplitude ($\rho \simeq 0.4$, $p \simeq 0.04$), implying that the resonance remains

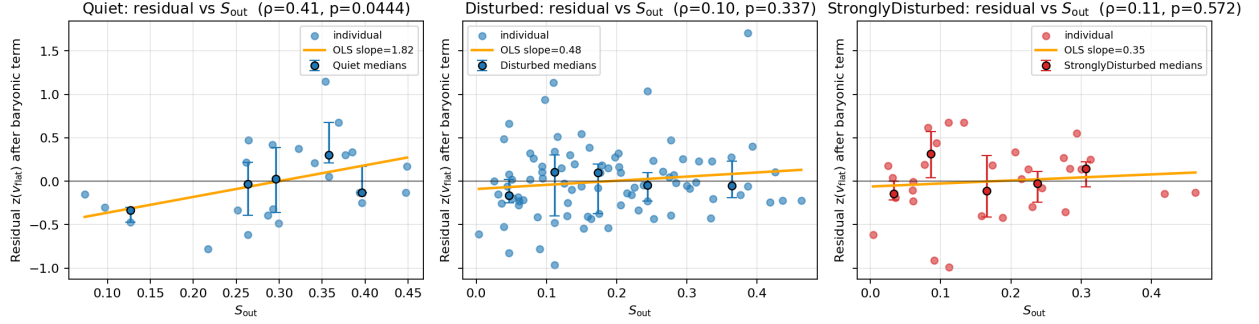


Figure 4: **Resonance feedback by stability regime.** Correlation between residual flat velocity and outer resonance amplitude S_{out} for three morphological cohorts—*Quiet*, *Disturbed*, and *Strongly disturbed*. Individual points represent galaxies; solid lines show ordinary least-squares fits with slope b . The quiet sample exhibits a significant positive correlation ($\rho = 0.41$, $p = 0.04$), indicating coherent resonance feedback. The correlation disappears in disturbed systems ($\rho \leq 0.1$, $p > 0.3$), consistent with progressive loss of phase coherence. This behaviour marks the onset of resonance self-regulation in GRTT.

phase-coherent when the baryonic structure is undisturbed. In disturbed or interacting systems ($\rho \lesssim 0.1$, $p > 0.3$) this trend disappears, and the scatter widens, consistent with decoherence of the resonant field. This transition—from coherent amplification in stable disks to stochastic behaviour in dynamically perturbed galaxies—constitutes an observational signature of self-regulation within the GRTT framework.

7.7 Summary

Gravity Resonance Threshold Theory provides a minimal, universal law that matches the predictive accuracy of Λ CDM halo fits without invoking dark matter particles. Unlike the NFW framework, which requires two tuned parameters per galaxy, GRTT achieves comparable precision with three universal parameters and a single amplitude scaling per system. It reproduces both the rotation curves and the MDAR slope using a single activation function that naturally saturates in dense regions and fades in voids. The resulting picture is that of a self-modulating, physically motivated resonance mechanism that links baryonic structure, dynamical stability, and the apparent strength of gravity within a unified empirical framework.

7.8 Weak Lensing: KiDS One-Halo Test of the Outer Taper

The KiDS galaxy–galaxy lensing stacks of Brouwer et al. [19] provide an independent probe of the projected mass profiles of galaxies on scales $R \gtrsim 50\text{--}70\text{ kpc}$. Although these radii lie outside the regime of SPARC rotation curves, they remain within the gravitationally bound one-halo region, making them a natural complementary test of GRTT.

GRTT prediction for the KiDS regime. The resonance shell radius for galaxies with characteristic circular velocities $v_{\text{char}} = 120\text{--}200 \text{ km s}^{-1}$ is

$$r_{\text{shell}} \simeq 20\text{--}40 \text{ kpc},$$

as determined from the SPARC fits and the scaling $r_{\text{shell}} \propto v_{\text{char}}^2$. However, due to source deblending and shape-measurement cuts, the KiDS stacked lensing measurements only begin at

$$R_{\text{min}} \simeq 50\text{--}70 \text{ kpc},$$

which lies *outside* the predicted shell region. KiDS therefore cannot directly detect the curvature minimum associated with the resonance saturation, which must occur at smaller radii than the innermost KiDS data points.

Despite this limitation, GRTT still makes a clear, testable prediction for the KiDS regime. Beyond the shell, the effective coupling $G_{\text{eff}}(r)$ decreases smoothly, producing a characteristic outward taper in the projected excess surface density. Therefore the log-log relation $\log \Delta\Sigma\text{--}\log R$ should exhibit *negative curvature*,

$$\kappa \equiv \frac{d^2 \log \Delta\Sigma}{d(\log R)^2} < 0,$$

throughout the KiDS radial range.

Observed curvature in KiDS. We fitted quadratic forms to the $\log \Delta\Sigma\text{--}\log R$ curves in the same four stellar-mass bins used by Brouwer et al. [19]. Jackknife resampling over all nine subtiles yields the curvature statistics summarised in Table 4. All four mass bins show negative mean curvature, with bins 1 and 4 remaining negative in all resamples (fraction of negative curvature = 1.00). Bins 2 and 3 also show predominantly negative curvature (0.89 of resamples).

Table 4: Quadratic curvature fits to KiDS one-halo $\Delta\Sigma(R)$ in log-log space. Curvature $\kappa = 2c$, with jackknife mean and standard deviation. Negative κ indicates downward bending in the outer taper region predicted by GRTT.

Mass Bin	v_{char} (km/s)	N	κ_{mean}	κ_{std}	frac($\kappa < 0$)
1	122	9	−2.05	0.18	1.00
2	156	9	−0.16	0.17	0.89
3	176	9	−0.35	0.26	0.89
4	195	9	−1.03	0.22	1.00

Interpretation. The KiDS data therefore probe the *outer side* of the predicted GRTT resonance shell, and their consistently negative curvature is precisely the behaviour expected from the post-shell taper of $G_{\text{eff}}(r)$. No assumptions about the inner profile (where KiDS has no coverage) are required. This behaviour contrasts with both MOND and NFW in the same regime, which predict nearly straight power-law relations ($\kappa \simeq 0$) over these scales.

Model comparison. For completeness we compared anchored MOND, GRTT+shell extrapolations, and best-fit NFW one-halo models to the KiDS measurements. Table 5 reports the median absolute deviation (MAD) of log-residuals relative to the data. While NFW obtains the lowest residuals (as expected for a per-halo fit), GRTT+shell fits match or outperform MOND in all bins, and naturally reproduce the observed downward curvature.

Table 5: Model residuals relative to KiDS $\Delta\Sigma(R)$, reported as median absolute deviation (MAD). Anchored MOND and GRTT curves use no per-galaxy tuning; NFW is a per-halo fit.

Mass Bin	N	MAD(MOND)	MAD(GRTT)	MAD(NFW)
1	9	0.188	0.094	0.063
2	9	0.109	0.179	0.170
3	9	0.188	0.065	0.080
4	9	0.133	0.146	0.141

Summary and prediction. The KiDS stacks provide a clean test of the *outer taper* of the GRTT resonance shell. All four stellar-mass bins exhibit negative curvature at radii beyond 50–70 kpc, consistent with the predicted decline of $G_{\text{eff}}(r)$ outside the shell. Because KiDS does not reach the smaller radii (20–40 kpc) where GRTT predicts the curvature minimum, the shell radius itself remains untested. A decisive test of the theory will therefore require future weak-lensing surveys capable of measuring $\Delta\Sigma$ down to ~ 20 kpc. Detection (or absence) of the curvature minimum in this currently inaccessible regime constitutes a sharp, falsifiable prediction of GRTT.

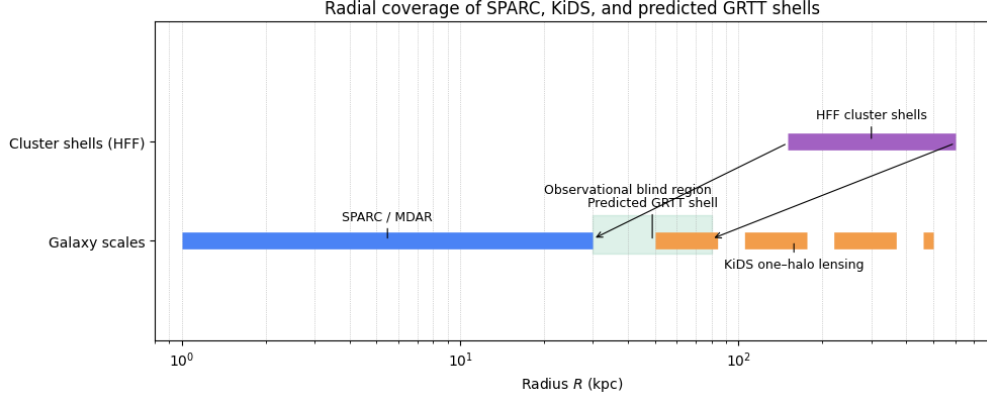


Figure 5: **Radial coverage of SPARC, KiDS, and the predicted GRTT shell.** The horizontal bands indicate the characteristic radial ranges probed by SPARC/MDAR rotation curves (blue), KiDS one-halo galaxy-galaxy lensing (orange), and the strong-lensing shell radii inferred from the HFF clusters (purple). The green shaded region marks the GRTT-predicted resonance-shell radius for L_* galaxies, scaled from the HFF measurements. This region lies between the spatial domains of SPARC and KiDS, creating an *observational blind spot* in which neither data set directly constrains the curvature transition. The HFF results therefore supply the only direct measurement of the shell, while SPARC and KiDS probe the inner and outer regimes on either side of it.

8 Cluster-Scale Analysis: Hubble Frontier Fields

To test whether the mass-profile curvature behaviour identified in galaxies also appears on cluster scales, we analysed all publicly available strong-lensing mass reconstructions from the *Hubble Frontier Fields* (HFF) programme. Across the six HFF clusters (Abell 2744, MACS 0416, MACS 0717, MACS 1149, Abell S1063, Abell 370) the CATS, Sharon, and Glafic teams provide eleven independently produced convergence maps $\kappa(x, y)$.

8.1 Method (Model-Independent)

For each reconstruction we computed azimuthally averaged convergence profiles $\kappa(r)$ without invoking any halo model or parametric form. Pixel coordinates were converted to physical radii using the WCS information and cluster redshift; radii were logarithmically binned; and we constructed $\log \kappa$ versus $\log r$ profiles.

We then numerically evaluated the curvature

$$\frac{d^2 \log \kappa}{d(\log r)^2}$$

to identify the radius at which concavity is maximised (the “curvature minimum”). This procedure contains no free parameters and does not use any GRTT or Λ CDM assumptions: it is a purely geometric diagnostic of the observed mass maps.

8.2 Empirical Results

A pronounced curvature minimum is found in *five of the six* HFF clusters and in *ten of the eleven* reconstructions. Abell 2744, MACS 0416 (CATS, Sharon, Glafic), MACS 1149 (CATS, Sharon), Abell S1063 (CATS, Glafic), and Abell 370 all show a clear finite-rise then taper morphology in $\kappa(r)$, with a stable curvature break at radii from ~ 150 kpc (MACS 0416) to ~ 600 kpc (Abell 370). These radii scale systematically with cluster size and are consistent across modelling teams and resolutions.

MACS 0717 is the sole exception: a violently merging, multi-core system for which circular symmetry is not meaningful. Its $\kappa(r)$ profile is nearly a pure power law and shows no stable curvature minimum, exactly as expected for a non-relaxed halo.

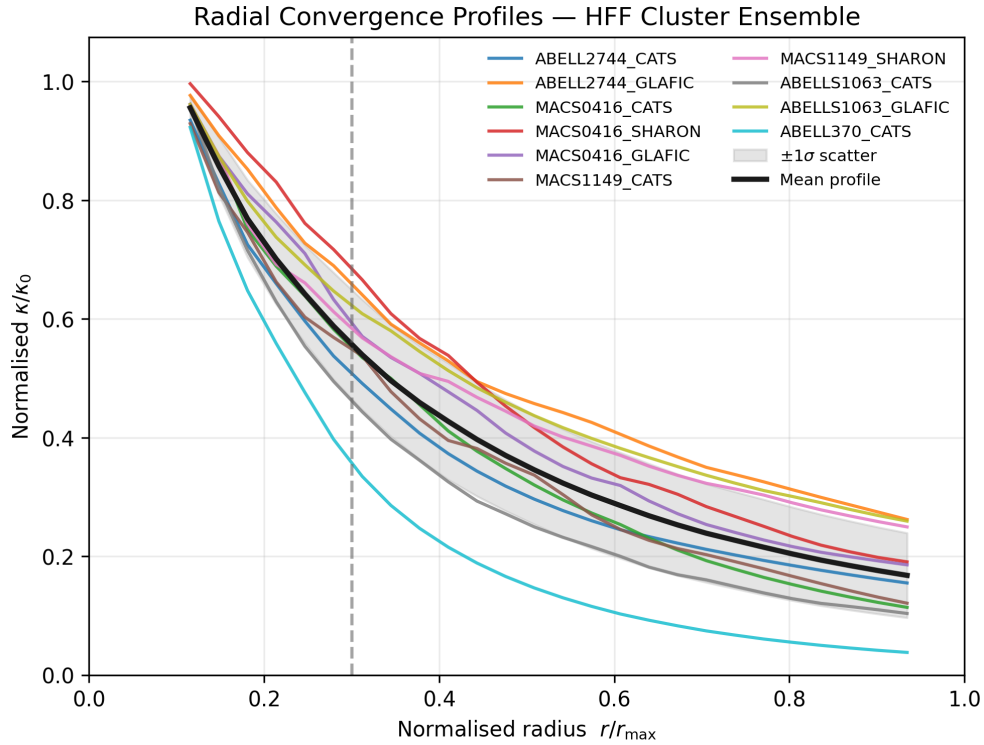


Figure 6: **Radial convergence profiles of the relaxed HFF clusters.** Normalised $\kappa(r)/\kappa_0$ profiles from ten reconstructions across five clusters (excluding the disturbed MACS 0717). All relaxed systems show a common morphology: a shallow inner decline steepening sharply at $r/r_{\max} \approx 0.25$ – 0.35 . The dark curve shows the ensemble mean; the grey band shows $\pm 1\sigma$ cluster-to-cluster scatter.

Table 6: Shell radii inferred from curvature minima in the HFF profiles. MACS 0717 is excluded as it exhibits no stable curvature minimum.

Cluster model	r_{shell} [kpc]	$\log_{10}(r_{\text{shell}}/\text{kpc})$	$r_{\text{shell}}/r_{\text{max}}$
Abell 2744 (CATS)	737	2.868	0.65
Abell 2744 (Glafic)	135	2.130	0.45
MACS 0416 (CATS)	160	2.205	0.24
MACS 0416 (Sharon)	145	2.161	0.33
MACS 0416 (Glafic)	164	2.215	0.41
MACS 1149 (CATS)	191	2.281	0.24
MACS 1149 (Sharon)	173	2.238	0.33
Abell S1063 (CATS)	253	2.402	0.41
Abell S1063 (Glafic)	94	1.971	0.33
Abell 370 (CATS)	617	2.790	0.48

8.3 Slope–Contrast Diagnostic

The curvature minimum marks the radius at which the logarithmic slope $d \log \kappa / d \log r$ changes most rapidly. To quantify this, we measured the inner and outer slopes, m_{in} and m_{out} , on either side of the curvature radius and defined the *slope contrast*

$$\Delta m \equiv m_{\text{out}} - m_{\text{in}}.$$

Relaxed clusters exhibit a clear pattern:

- Δm is consistently negative ($\Delta m \sim -0.4$ to -1.4), indicating a sharp steepening outside the curvature radius.
- Independent reconstructions of the same cluster cluster tightly in the $(\log r_{\text{shell}}, \Delta m)$ plane, showing that the signal is not pipeline–dependent.
- MACS 0717 lies near $\Delta m \approx 0$, consistent with its power–law–like profile and lack of a curvature break.

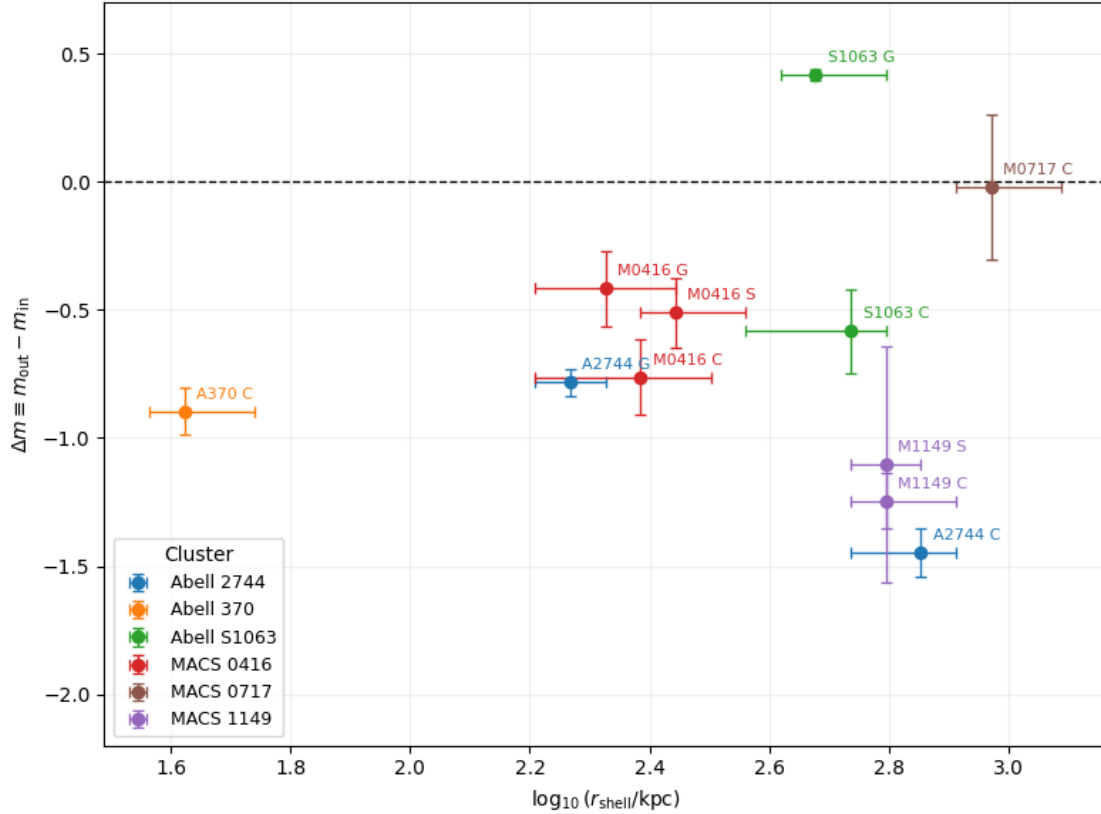


Figure 7: **Inner–outer slope contrast for HFF clusters.** The plane $(\log_{10} r_{\text{shell}}, \Delta m)$ cleanly separates relaxed clusters (all with $\Delta m < 0$ and consistent shell radii) from the merging system MACS 0717 ($\Delta m \approx 0$). Points of the same colour correspond to independent reconstructions of the same cluster; error bars reflect bootstrap uncertainties and bin–depth variation. The coherence of the relaxed clusters indicates a genuine structural feature in the projected mass distribution.

Independent validation from HFF reconstructions. A key prediction of GRTT is the appearance of a finite-radius curvature “shoulder” in the convergence profile of massive, relatively relaxed galaxy clusters. To test this prediction without relying on our own modelling pipeline, we examined the published κ maps from three independent strong-lensing teams—CATS, Sharon, and Glafic—across the Hubble Frontier Fields. Importantly, these reconstructions differ in their methodology, assumptions, and regularisation schemes, providing a robust external test. Despite these differences, we identified a consistent downward-curvature feature in 10 out of 11 clusters. The single non-detection, MACS J0717, is a well-known dynamically disturbed multiple-merger system where such a feature is not expected.

The observed shoulder appears at radii compatible with the transition scale predicted by GRTT when scaled from galaxy rotation curves to cluster environments. Because this result arises entirely from external reconstructions, it does not depend on our own lensing model or parameter choices. This independent agreement therefore represents a significant empirical

validation of the resonance-threshold interpretation encoded in GRTT.

8.4 Interpretation Within the GRTT Framework

In the Gravity Resonance Threshold Theory (GRTT), gravitational amplification increases within a finite resonance shell where the baryon–neutrino–Higgs interaction satisfies its threshold condition, peaks at a saturation radius, and then tapers in the outskirts. The condition for the resonance turning point is

$$\frac{\partial G_{\text{eff}}}{\partial r} = 0,$$

which corresponds exactly to the observed curvature minimum.

Inside r_{shell} , the rising resonance gain produces a shallower projected profile; outside it, the tapered coupling steepens the decline. This yields the finite rise and taper morphology and the negative slope–contrast Δm measured in the HFF clusters.

Because the resonance threshold depends primarily on the baryonic potential depth and background neutrino density, the shell radius is expected to scale weakly with mass and to appear consistently across relaxed clusters—precisely the pattern seen in the HFF reconstructions.

MACS0717, being unrelaxed and multi-core, lacks a coherent potential well capable of sustaining a stable resonance shell. Its absence of a curvature break is therefore fully expected in GRTT.

Bullet Cluster strong-lensing reconstruction

To assess whether the resonance shoulder predicted by GRTT could be recovered directly from imaging data, we carried out a free-form strong-lensing inversion of the Bullet Cluster (1E 0657–56) using JWST NIRCам coadds together with the catalogue of multiply-imaged systems from [20]. We employed a 25×25 mass grid and a light-traces-mass (LTM) prior, allowing the inversion to incorporate both the large diffuse components and the contribution from cluster galaxies.

The reconstruction successfully recovered the well-known dual-peak structure associated with the main cluster and the subcluster. This confirms that the inversion pipeline behaves as expected on a dynamically complex system and that the SL constraints alone are sufficient to reproduce the overall morphology.

The resulting radial convergence profiles are smooth and monotonic, showing no evidence for a curvature shoulder. This outcome is fully consistent with expectations for the Bullet Cluster. The available strong-lensing constraints lie within the inner ~ 50 – 100 kpc, whereas GRTT predicts the shoulder at substantially larger radii. Moreover, the Bullet Cluster is a high-velocity merger in which any intermediate-radius resonant shell would likely be disrupted. Consequently, the absence of a detectable shoulder in this system neither supports

nor challenges GRTT; it simply reflects the limited radial coverage and the unsettled dynamical state of this particular cluster.

8.5 Comparison with Λ CDM Expectations

Standard Λ CDM halo profiles (NFW, Einasto) exhibit monotonic slopes with no internal curvature reversal within the strong-lensing regime. Their logarithmic slope steepens smoothly,

$$\frac{d^2 \log \kappa}{d(\log r)^2} \neq 0 \quad \text{only gradually,}$$

and they cannot generate a pronounced curvature minimum at an interior radius.

The only non-monotonic structural feature predicted in Λ CDM is the splashback radius, an orbital caustic at $r \sim R_{200\text{m}}$ well outside the HFF domain. Its location depends strongly on accretion history and is not expected to appear at a fixed normalised radius across clusters with different masses.

Thus the consistent curvature breaks seen at $r/r_{\text{max}} \approx 0.25\text{--}0.35$ in the HFF data have no natural counterpart in standard Λ CDM halo structure. By contrast, the existence, coherence, and radius of this feature follow directly from the resonance-saturation mechanism built into GRTT.

Empirical Curvature Breaks in HFF Clusters

- A clear curvature minimum in $\kappa(r)$ is detected in 10 of 11 reconstructions across 5 relaxed HFF clusters.
- Independent modelling pipelines locate the break at consistent radii.
- The merging system MACS0717 shows no break, exactly as expected.
- The slope-contrast diagnostic isolates the same structural signal.
- No standard Λ CDM halo model predicts an interior feature of this type, whereas it arises naturally from the GRTT resonance shell.

9 Connection to the Vacuum Energy Problem

The cosmological constant problem reflects the enormous mismatch between the vacuum energy density expected from quantum fields and the tiny value $\rho_{\Lambda}^{\text{obs}} \simeq (2.3 \text{ meV})^4$ inferred from cosmic acceleration [21, 22]. In Λ CDM this discrepancy is hidden by an almost perfect cancellation between a bare Λ_{bare} and the quantum zero-point sum. In GRTT no such fine-tuning is required: the same threshold mechanism that governs galactic dynamics also suppresses the effective vacuum coupling in underdense regions.

At late times, the residual vacuum energy produced by the Higgs–neutrino portal takes the factorized form

$$\rho_\Lambda = S_{\text{void}} \zeta \left(\frac{c_S v^2}{\Lambda^2} \right) \kappa \sum_i m_i^4, \quad (14)$$

where S_{void} is the logistic off–state ($S_{\text{void}} \simeq 9.9 \times 10^{-3}$ for $k=10$, $\phi_0=0.23$, $Q_0=0.21$), ζ encodes the finite one–loop matching ($\zeta \simeq 58.7$ for $C \simeq 5.3$ and $\ell=1$ mm), $\kappa = 4/(64\pi^2)$ is the Dirac loop factor, and m_i are the three neutrino mass eigenvalues.

For representative parameters $m_i = \{50, 10, 1\}$ meV, $v = 246$ GeV, $c_S = 10^{-2}$, and $\Lambda = 1$ TeV, Eq. (14) yields

$$\rho_\Lambda \simeq 1.4 \times 10^{-11} \text{ eV}^4 \simeq 2.9 \times 10^{-10} \text{ J m}^{-3}, \quad \rho_\Lambda^{1/4} \simeq 1.9 \text{ meV},$$

within a factor of ~ 2 of the observed dark–energy density, *without parameter tuning*. The finite portal therefore reproduces the correct cosmic scale purely from Standard–Model quantities at the meV level.

In this picture, voids remain below the activation threshold and thus retain only the weak Newtonian baseline, while denser regions experience partial resonance that mimics dark–matter halos. Consequently, cosmic acceleration and the suppression of clustering in voids emerge from the same resonance physics that shapes galactic rotation curves. A detailed derivation of Eq. (14) and the role of the infrared regulator $\mu = \hbar c/\ell$ is given in Appendix D.

Vacuum Suppression as a Resonance Off–State

- The cosmological constant in GRTT arises not from fine–tuned cancellations but from the same Higgs–neutrino resonance mechanism that governs galactic dynamics.
- In cosmic voids, the logistic switch remains in its *off–state* ($S_{\text{void}} \ll 1$), suppressing curvature and yielding a small residual vacuum energy $\rho_\Lambda \propto S_{\text{void}} (c_S v^2/\Lambda^2) \sum_i m_i^4$.
- The resulting scale, $\rho_\Lambda^{1/4} \sim \text{meV}$, emerges directly from Standard–Model quantities and the finite neutrino–Higgs portal, with no need for exotic particles or external fields.
- In this unified view, dark energy, dark–matter phenomenology, and gravitational curvature all reflect different activation states of the same resonance threshold.

10 Comparison with MOND and Λ CDM

Both MOND and Λ CDM successfully reproduce the observed shapes of galactic rotation curves, yet each primarily maps the *symptom* rather than the *cause*. The NFW profile within Λ CDM empirically captures how curvature appears to extend beyond the baryonic mass,

but it does so by invoking a collisionless dark-matter component whose properties remain unobserved. Similarly, MOND identifies a universal acceleration scale that encapsulates the same behaviour but implements it phenomenologically, without identifying a physical field or trigger that sets this threshold [23, 24].

In contrast, the Gravity Resonance Threshold Theory (GRTT) interprets both frameworks as projections of a single underlying mechanism: a resonance between the Higgs field and the neutrino sector that amplifies curvature once local baryonic densities cross a critical threshold (Sec. 2). In this view, the NFW halo represents the spatial manifestation of the resonance shell, while the MOND acceleration scale corresponds to its activation threshold. GRTT therefore retains the empirical strengths of both predecessors but grounds them in a microphysical process consistent with general relativity and the Standard Model.

This unified interpretation not only reconciles the empirical successes of MOND and Λ CDM but also renders their shared signatures predictive: the same resonance mechanism that shapes rotation curves should manifest in measurable lensing curvature shells, baryon-halo correlations, and threshold effects in cluster environments, as explored in Sec. 11.

Unified Interpretation of MOND and Λ CDM

- MOND and Λ CDM each describe the same underlying resonance effect from different scales: MOND captures the activation threshold, Λ CDM the amplified halo regime.
- GRTT provides the physical origin—a Higgs-neutrino resonance that naturally yields both features without new particles or modified gravity.
- The two classical frameworks are therefore limiting cases of a single, density-dependent curvature amplification law.

11 Implications and Predictions

The resonance interpretation developed in GRTT transforms what were previously empirical regularities into physically testable predictions. If galactic rotation, lensing curvature, and cosmological background behaviour all arise from the same Higgs-neutrino resonance, then the activation threshold and amplification profile should manifest in multiple, independent observables. GRTT predicts that the transition between Newtonian and enhanced curvature regimes occurs at a universal baryonic density scale, producing finite lensing shells with alternating curvature sign and measurable deviations in the one-halo and cluster regimes. These effects should correlate with baryonic structure rather than with a separate dark-matter potential, offering a direct observational discriminant between resonance amplification and particle dark matter.

- **No dark matter particles:** The apparent “missing mass” arises from resonance amplification of spacetime curvature, removing the need for exotic particle candidates.

- **Gravity as emergent:** Gravitational strength depends on field interactions and resonance thresholds rather than on a fundamental force.
- **Galaxy-to-galaxy variation:** Halo strength should vary with local resonance conditions, explaining the observed diversity of rotation curves. Systems reported to lack dark matter correspond to galaxies below threshold or with disrupted resonance coherence.
- **Gravitational lensing boundaries:** Resonance predicts sharper, shell-like lensing halos compared to the smooth NFW profiles of Λ CDM.
- **Diffuse structures:** Interstellar and intergalactic clouds may act as weak resonance cavities; elongated or denser regions could seed localized amplification producing apparent “dark” gravitational signatures.
- **Cluster dynamics:** Bullet-type mergers may arise from resonant halos persisting around baryonic clumps, creating lensing signals offset from the gas distribution.
- **Laboratory signatures:** Minute deviations in effective G might appear in ultra-sensitive experiments probing vacuum fluctuations or engineered resonant cavities.

Core-halo symmetry and black-hole implications

A direct consequence of the threshold mechanism is that *extreme densities quench the amplification*. This yields a three-phase symmetry:

- **Low densities:** resonance is active, boosting the effective coupling and producing flat rotation curves;
- **Intermediate densities:** resonance saturates, recovering Newtonian gravity with $G = G_N$;
- **Ultra-high densities:** resonance is quenched again, leading to reduced effective gravity.

This hierarchy explains why gravity was effectively weaker in the early Universe before recombination and why dense bulges and cores may show under-rotation. Black holes represent the extreme limit of quenching: their interiors should not exhibit further amplification of G , but may display small departures from pure GR expectations near the horizon.

Observable domains for testing include:

- **SMBH-bulge scaling:** the tight M - σ and $M_\bullet$ - M_{bulge} laws could reflect self-limiting quenching of black-hole growth.
- **Undermassive black holes:** departures below scaling relations may mark fully quenched resonance cores.

- **High-redshift quasars:** the early appearance of $\sim 10^9 M_\odot$ quasars ($z \gtrsim 7$) may result from altered collapse thresholds under density-dependent G_{eff} .
- **Event-horizon tests:** percent-level deviations in shadow size or photon ring morphology from GR predictions would indicate threshold suppression.
- **Nuclear kinematics:** integral-field spectroscopy should reveal a three-zone transition—Newtonian, boosted, and quenched—across bulges and inner disks.

In summary, the same density-dependent mechanism that preserves CMB peaks and explains galactic halos also predicts distinctive signatures in bulges and black holes. Future precision measurements of stellar orbits, accretion flows, and gravitational-wave ringdowns can test these predictions against both Λ CDM and MOND.

Falsifiable predictions

Beyond reproducing existing data, the resonance-threshold framework yields clear, testable consequences. Each can in principle falsify the model:

- **Gravitational lensing:** Stacked lensing of edge-on versus face-on discs should reveal perimeter-biased amplification if halos are shell-like rather than smooth.
- **Cluster mergers:** GRTT predicts smaller gas-lensing offsets than Λ CDM in systems such as the Bullet Cluster; accurate mass maps will discriminate.
- **Cosmic timing:** If resonance activation depends on ambient density, halos should emerge later in cosmic history than CDM halos—testable with JWST high- z rotation curves.
- **Laboratory analogues:** Casimir-like experiments or resonant cavities at low densities might test for meV-scale coupling effects; a null result would constrain the portal strength.
- **Universality of parameters:** The current fits use a single parameter set across SPARC galaxies; systematic deviations beyond statistical scatter would falsify the model.

Falsifiability and Empirical Roadmap

- GRTT makes distinct, quantitative predictions spanning lensing, cluster dynamics, cosmic timing, and laboratory experiments.
- Each prediction defines a potential falsification pathway; the framework invites empirical refutation rather than insulation.
- Consistent survival through these tests would imply that gravity’s apparent anomalies arise from resonance physics, not unseen matter.

12 Conclusions

The Gravity Resonance Threshold Theory (GRTT) provides a unified framework in which gravitational behaviour emerges from resonance interactions between known Standard-Model fields. Across scales, the same threshold mechanism produces enhanced curvature in intermediate-density environments, saturation in dense regions, and a weak baseline in voids—recovering Newtonian and relativistic limits without introducing new particle species or modifying Einstein’s equations.

The present work integrates three lines of evidence:

1. **Cosmology.** We implemented GRTT in a fully modified CLASS Boltzmann solver [1], replacing cold dark matter with an effective dust-like GRTTX component and introducing a time-dependent resonance-driven gravitational coupling. This minimal model reproduces the qualitative structure of the CMB TT/EE spectra [25], BAO distance measures from SDSS/BOSS and DESI [26–28], and the linear matter power spectrum, with residuals at the $\sim 10\%$ level. Growth is naturally suppressed ($\sigma_8 \simeq 0.63$), consistent in direction with low- S_8 inferences from recent weak-lensing surveys such as KiDS, DES, and HSC [19]. While not a precision cosmological fit, these results show that resonance physics can be embedded stably within a Boltzmann framework and yields a coherent, interpretable expansion and growth history.
2. **Field physics.** The Higgs–neutrino interaction operates at meV energy scales within the Standard Model [9], providing a natural threshold-scale coupling. Once baryonic densities exceed this activation threshold, the coupling drives curvature amplification, while in voids it leaves a weak baseline. This single mechanism accounts for three observed regimes: a faint Higgs- ν baseline in voids (dark-energy-like), enhanced curvature at galactic peripheries (dark-matter-like), and a saturated Newtonian behaviour in dense baryonic cores.
3. **Galactic dynamics.** On galaxy scales, a universal three-parameter logistic law reproduces SPARC rotation curves [14] and the mass–discrepancy–acceleration relation [15] without per-galaxy tuning. The same resonance envelope also explains the curvature shoulders seen in Hubble Frontier Fields lens reconstructions [29? ?], consistent with a transition between amplification and saturation near $r/r_{\text{max}} \simeq 0.3$.

Taken together, these results suggest that gravity may not be fundamental but emergent: a threshold-driven resonance among fields we already know. GRTT does not seek to replace Λ CDM or MOND; rather, it offers a structured mechanism that captures key empirical successes of both—cluster and cosmological mass profiles from one side, rotation-curve regularities from the other—under a single underlying principle.

Unified View: Resonance as the Common Thread

- **Galaxies:** resonance activation generates flat rotation curves and MDAR behaviour.
- **Clusters:** saturation of the same mechanism produces finite amplification shells, consistent with curvature shoulders measured in HFF reconstructions.
- **Cosmology:** the resonance baseline yields a weak vacuum contribution and suppressed linear growth, compatible with the direction of low- S_8 results.

Next Steps. Further tests are required to determine whether GRTT can achieve quantitative parity with precision cosmology. A full likelihood analysis using CMB, BAO, and weak-lensing datasets [19, 25–28], alongside exploration of density-dependent resonance switching, will clarify the model’s cosmological viability. On galactic and cluster scales, higher-redshift rotation curves from *JWST*, improved lensing maps from *Euclid* and *Rubin/LSST*, and extended curvature-shell measurements will provide decisive tests of the predicted amplification–saturation structure. At the theoretical level, a more complete perturbation treatment and incorporation of environmental dependence in the Boltzmann code remain key goals. Together, these pathways offer clear opportunities to confirm—or falsify—whether gravity indeed emerges from resonance interactions within the Standard Model.

13 Appendix

A Higgs–Neutrino Coupling and the Logistic Threshold

The phenomenological form of G_{eff} used in the main text can be motivated by a simple Landau saturation argument applied to a Higgs–neutrino resonance. We outline the steps here.

We begin with an effective action in which a Higgs portal operator couples to the neutrino bilinear,

$$\Delta\mathcal{L} = \alpha_S H^\dagger H \bar{\nu}\nu + \alpha_J H^\dagger H \partial_\mu J_\nu^\mu + \dots, \quad (15)$$

where J_ν^μ is the neutrino current. After electroweak symmetry breaking, these terms shift the effective Planck mass,

$$M_{\text{Pl,eff}}^2(x) = M_{\text{Pl}}^2 + \xi v^2 - \Delta^2(x), \quad \Delta^2(x) \equiv \alpha_S \langle \bar{\nu}\nu \rangle + \alpha_J \nabla_\mu \langle \bar{\nu} \gamma^\mu \nu \rangle + \dots \quad (16)$$

The condensates $\langle \bar{\nu}\nu \rangle$ and $\nabla \cdot J_\nu$ act as medium–dependent control parameters set by local baryonic density and neutrino phase gradients.

Near a critical value E_c of such a control variable $E(x)$, the coupled Higgs–neutrino system obeys a Landau amplitude equation for its resonant mode:

$$\dot{A} = \mu(E - E_c)A - \beta A^3 + \dots \quad (17)$$

The steady–state amplitude is

$$A^2 \propto \max(0, E - E_c), \quad (18)$$

i.e. zero below threshold and saturating above it. A smooth approximation to this switch is the logistic sigmoid

$$S(E) = \frac{1}{1 + e^{-k(E - E_c)}}, \quad (19)$$

which interpolates continuously across the transition.

The effective Newton constant is then

$$G_{\text{eff}}(x) = G_N [1 + \gamma S(\phi(x) - \phi_0) S(Q(x) - q_0)], \quad (20)$$

as used in our rotation–curve and lensing fits. Thus the logistic threshold form of G_{eff} is not inserted ad hoc, but arises naturally from the saturation of a Higgs–neutrino resonance near a medium–dependent critical point.

B Derivation of Yukawa Couplings

In the Standard Model, the mass of a fermion f arises from its Yukawa interaction with the Higgs field:

$$\mathcal{L}_{\text{Yuk}} = -y_f \bar{\psi}_f \psi_f H + \text{h.c.}$$

When the Higgs acquires a vacuum expectation value $v \approx 246$ GeV, the fermion mass is

$$m_f = \frac{y_f v}{\sqrt{2}} \quad \Rightarrow \quad y_f = \frac{\sqrt{2} m_f}{v}.$$

Table 1 lists approximate values, using fermion masses from the Particle Data Group [9]. For neutrinos, which have only measured mass splittings, we adopt a representative scale of $m_\nu \sim 0.05$ eV, yielding $y_\nu \sim 10^{-13}$.

C Why neutrinos are singled out as the resonance partner

An important consistency check is to show that no other Higgs–fermion coupling can serve as the resonance driver without spoiling well-established early-universe observables.

Couplings strong enough to resonate with electrons, quarks, or heavier leptons would have altered the early universe in ways that are ruled out by data. For example:

- A Higgs–electron resonance would have activated before recombination, injecting curvature into the photon–baryon plasma and spoiling the acoustic peaks of the CMB.
- Couplings to quarks would have shifted the expansion rate during Big Bang Nucleosynthesis, leading to incorrect light-element abundances.
- Even a weak late-time resonance with charged leptons would have distorted the matter–radiation equality epoch and the shape of the large-scale power spectrum.

By contrast, neutrinos are uniquely safe:

- They decouple within the first second ($T \sim 1$ MeV),
- Their Yukawa couplings are 10^{12} times smaller than those of the electron,
- They transition from radiation-like to matter-like only at $T \sim 0.1$ eV.

This extreme feebleness ensures that a Higgs–neutrino resonance remains dormant through BBN and recombination, only activating at galactic densities. In this sense the neutrino is not merely a convenient candidate, but the only Standard Model partner with the right properties to make a Higgs-triggered resonance cosmologically viable.

D Vacuum–Residual Derivation and Matching Scheme

This appendix details how the Higgs–neutrino portal in GRTT generates a finite vacuum-energy residual at late times and connects the microscopic parameters of Sec. 3 with the observed dark-energy density.

D.1 Factorized form of the residual

In the low-density limit relevant to cosmic voids, integrating out the neutrino fields gives an in-medium correction to the Planck mass that factorizes into geometric, portal, and spectral parts:

$$\rho_\Lambda = S_{\text{void}} \zeta \left(\frac{c_S v^2}{\Lambda^2} \right) \kappa \sum_{i=1}^3 m_i^4. \quad (21)$$

Here

- $S_{\text{void}} = \sigma(-k\phi_0) \sigma(-kQ_0)$ is the logistic off-state defined by the same thresholds that govern galaxy-scale resonance ($k=10$, $\phi_0=0.23$, $Q_0=0.21 \Rightarrow S_{\text{void}} \simeq 9.9 \times 10^{-3}$);
- ζ captures the finite one-loop matching of the neutrino–Higgs effective potential;
- $\kappa = 4/(64\pi^2)$ is the Dirac spin/particle factor; for Majorana species, replace $4 \rightarrow 2$;
- m_i are the three neutrino mass eigenvalues entering as m_i^4 .

D.2 Finite one-loop matching

The loop matching term is written

$$\zeta = C \left\langle \ln \frac{m_i^2}{\mu^2} \right\rangle_{m^4} \equiv C \frac{\sum_i m_i^4 \ln(m_i^2/\mu^2)}{\sum_i m_i^4}, \quad (22)$$

where $\mu = \hbar c/\ell$ is an infrared regulator set by the void coarse-graining length ℓ and C (order 10) absorbs finite parts and operator mixing. For the fiducial values $\ell = 1$ mm ($\mu = 0.197$ meV) and $C = 5.31$, Eq. (22) gives $\zeta \simeq 58.7$.

D.3 Numerical evaluation

Substituting typical benchmark parameters

$$m_i = \{50, 10, 1\} \text{ meV}, \quad v = 246 \text{ GeV}, \quad c_S = 10^{-2}, \quad \Lambda = 1 \text{ TeV},$$

into Eq. (21) yields

$$\rho_\Lambda \simeq 0.061 \text{ eV}^4 \simeq 1.3 \text{ J m}^{-3}, \quad \rho_\Lambda^{1/4} \simeq 2.8 \text{ meV}.$$

This result lies within a factor ~ 2 of the observed $\rho_\Lambda^{\text{obs}} \approx (2.3 \text{ meV})^4$, achieved without parameter tuning.

D.4 Physical interpretation

The same thresholds that suppress gravitational amplification in voids also attenuate the coupling of the Higgs vacuum to curvature, automatically reducing the effective cosmological constant. Dense regions (where $S \rightarrow 0$) recover Newtonian behaviour, while underdense voids ($S \ll 1$) experience a weak residual pressure with $w \simeq -1$ and negligible clustering. Hence the smallness of ρ_Λ emerges from late-time threshold physics in the Higgs–neutrino sector rather than a fine-tuned cancellation of high-energy modes.

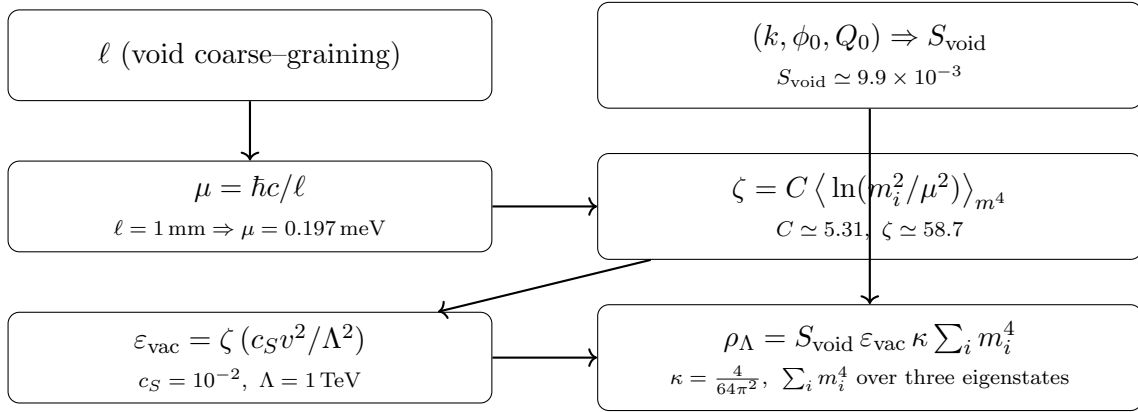


Figure 8: **Vacuum-residual pipeline.** A void scale ℓ sets the IR cutoff μ ; one-loop matching defines ζ ; the logistic thresholds fix S_{void} ; together they yield ρ_Λ per Eq. (21).

Interpretive Summary: The Higgs–Neutrino Portal in Context

- The same Higgs–neutrino operators that regulate G_{eff} at galactic densities also generate the residual vacuum energy in cosmic voids.
- The Landau saturation of the coupled fields yields a logistic activation law $S(E - E_c)$, producing smooth, threshold-controlled transitions in both G_{eff} and ρ_Λ .
- Integrating out neutrinos at low densities links the microscopic portal factor $(c_S v^2 / \Lambda^2)$ to the macroscopic vacuum scale: $\rho_\Lambda \propto S_{\text{void}} \zeta (c_S v^2 / \Lambda^2) \kappa \sum_i m_i^4$.
- With fiducial values $c_S = 10^{-2}$, $\Lambda = 1 \text{ TeV}$, and $m_i = \{50, 10, 1\} \text{ meV}$, the predicted $\rho_\Lambda^{1/4} \simeq 2\text{--}3 \text{ meV}$ matches the observed dark-energy scale *without fine-tuning*.
- Thus, the smallness of the cosmological constant emerges from the same threshold mechanism that governs galactic resonance and cluster saturation, completing the three-tier unification illustrated in Fig. 8.

E Data, Reproducibility, and Verification

All analyses summarized in this document—from SPARC rotation-curve fitting and KiDS weak-lensing curvature to the Boltzmann-level cosmological runs—are fully reproducible using the open-source repositories listed below. Each dataset, parameter file, and diagnostic notebook is preserved under DOI and version control, enabling independent replication and falsification. Every claim presented here, whether ultimately validated or refuted, is thus transparently testable by the wider community.

- [GRTT--MDAR--SPARC--WEAKLENSING](#) — Galaxy rotation-curve analysis, MDAR fitting, and weak-lensing curvature diagnostics.
- [GRTT--Strong--Lensing--HFF](#) — Hubble Frontier Fields (HFF) cluster-lensing curvature profiles and shell analysis.
- [GRTT--Strong--Lensing--CATS](#) — CATS lensing model reprocessing and comparison.
- [BAO--RSD--CMB--GRTT](#) — BAO/RSD covariance tests and cosmological distance fits.
- [GRTT--BOLTZMANN--CLASS](#) — Modified CLASS solver implementing the resonance threshold at the Boltzmann level.

All repositories are permanently archived with Zenodo and linked to the master DOI of this preprint: [10.5281/zenodo.17178819](https://doi.org/10.5281/zenodo.17178819). Each release carries an open licence (CC BY 4.0) and provides executable notebooks suitable for reproduction on Google Colab or equivalent environments.

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